

RESEARCH

Open Access



Machine learning-based hybrid technique to enhance cyber-attack perspective

Aun Abbas¹, Muqaddas Salahuddin², Muhammad Zohaib Khan³, Abdullah Ayub Khan⁴, Fahim Uz Zaman⁵, Syed Azeem Inam⁶, Ghadah Aldehim⁷, Tehseen Mazhar^{8*} and Muhammad Amir Khan^{9*}

Abstract

The rapid proliferation of interconnected devices and Internet of Things (IoT) applications has intensified cybersecurity risks for individuals, organizations, and critical infrastructures. Despite advances in encryption and authentication protocols, networks remain vulnerable to sophisticated cyber threats, particularly Distributed Denial of Service (DDoS) attacks. This study investigates the effectiveness of hybrid machine learning (ML) models in distinguishing between benign traffic and two DDoS variants—DDoS-ACK and DDoS-PSH-ACK. Building on recent advances in intrusion detection, we integrate Fuzzy C-Means clustering with classifiers such as Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), and AdaBoost (ADB). Experiments conducted on benchmark datasets evaluate models using accuracy, precision, recall, F1-score, and Matthews Correlation Coefficient (MCC). Results show that the hybrid models outperform standalone classifiers, particularly in handling imbalanced datasets and overlapping class boundaries. These findings underscore the practical viability of clustering-enhanced ML for real-time, attack-specific intrusion detection systems in IoT and cloud-enabled environments.

Keywords Multi-Layer perceptron (MLP), Cyber-Attacks detection (CADs), Support vector machine (SVM), Fuzzy C-Means, And adaboost (ADB) algorithms

*Correspondence:

Tehseen Mazhar
tehseenmazhar719@gmail.com
Muhammad Amir Khan
amirkhan@uitm.edu.my

¹Department of Computer Science and Cyber Security, Ravensbourne University, London, England

²Faculty of Computer Science and Information Technology, Superior University, Lahore 54000, Pakistan

³Department of Information Technology, Shaheed Mohtarma Benazir Bhutto Institute of Trauma, Karachi, Pakistan

⁴Department of Computer Science, Bahria University Karachi Campus, Karachi 75260, Pakistan

⁵Digital Technologies, Newcastle College University Centre (NCUC), Rye Hill Campus, Scotswood Rd, Tyne and Wear NE38, Newcastle, UK

⁶Department of Artificial Intelligence and Mathematical Sciences, Sindh Madressatul Islam University, Aiwan-e-Tijarat Road, Karachi 74000, Pakistan

⁷Department of Information Systems, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia

⁸School of Computer Science, National College of Business Administration and Economics, Lahore 54000, Pakistan

⁹School of Computing Sciences, Faculty of Computer Science and Mathematics, Universiti Teknologi MARA, Shah Alam, Selangor 40450, Malaysia



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Introduction

The rapid advancement of digital technologies and the widespread use of smart devices have transformed modern life. Smart systems comprising devices embedded with sensors and connected through networks are integral to sectors like healthcare, transportation, agriculture, and manufacturing. These systems offer significant advantages, such as automation, improved efficiency, and real-time data exchange. However, their interconnectivity also introduces new vulnerabilities, increasing the potential for cyberattacks. Cybersecurity has become an essential discipline focused on safeguarding digital systems and networks from malicious threats. Traditional security approaches, which often rely on manual rule-based methods, struggle to cope with the complexity and volume of modern cyber threats. The dynamic nature of today's cyber landscape, including the emergence of advanced persistent threats (APTs), phishing, ransomware, malware, and Distributed Denial of Service (DDoS) attacks, demands more adaptive and predictive defense mechanisms [1].

One of the most significant shifts in modern cybersecurity is the integration of Artificial Intelligence (AI) and Machine Learning (ML) techniques. These technologies enable systems to detect, predict, and respond to threats by learning from historical patterns and adapting to new attack strategies. ML algorithms are now widely used in network traffic analysis, intrusion detection systems (IDS), anomaly detection, and botnet detection. Their ability to automate threat detection, reduce human error, and operate in real-time makes them indispensable in modern security architectures [2, 3].

Among various threats, botnets pose a substantial risk. Botnets consist of a network of compromised devices controlled remotely by an attacker, or "botmaster." These networks are often used to execute coordinated attacks such as phishing, malware propagation, and DDoS attacks [4, 5]. Unlike typical malware, botnets communicate using structured command and control (C&C) channels ranging from IP-based systems to peer-to-peer (P2P) architectures making them more elusive and resilient [6, 7].

To counteract such threats, Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) are commonly employed, especially in Internet of Things (IoT) environments. However, traditional detection techniques suffer from limitations such as high false-positive rates, lack of adaptability, and dependence on labeled data. Recent research has proposed advanced ML and Deep Learning (DL) models including Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Convolutional Neural Networks (CNN) to improve detection capabilities. Techniques like Transfer Learning, Federated Learning, and attention mechanisms (e.g.,

CBAM) have also been introduced to enhance model accuracy, adaptability, and scalability. Figure 1 illustrates the attacks in the realm of cyber-security.

Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) are fundamental components for securing Internet of Things (IoT) applications and infrastructures [8]. As IoT ecosystems grow increasingly complex, anomaly detection systems have been proposed to address evolving threats. For example [8], proposed integrated incremental, transfer, and deep learning models to enhance detection performance, reduce costs, and adapt to concept drift—a key challenge in dynamic environments. Various detection techniques are reviewed in [9], including cooperative IoT strategies, hybrid models, anomaly-based, and signature-based systems, each offering unique strengths. Building on this [10], proposed collaborative approaches to improve resilience through device redundancy, ensuring continuity of service during attacks. Deep learning (DL) methods are particularly effective in identifying sophisticated attacks, such as those on 5G systems [11]. For instance [12], developed a neural network-based IDS for electric vehicle charging stations using LSTM and GRU, focusing on reducing false positives. Advanced models incorporating Convolutional Block Attention Modules (CBAM) and Transfer Learning (TL) have further improved classification accuracy by refining attention to spatial and channel features [13].

Network architecture plays a crucial role in IDS performance. Studies comparing EPA and MUD architectures demonstrate EPA's superiority in anomaly detection due to its holistic communication state analysis [14, 15]. Additionally [16], introduced an adaptive online framework for detecting distributed denial-of-service (DDoS) attacks while handling concept drift using techniques like ARE, KNN, ADWIN, and DDM. One of the main challenges in botnet and DDoS detection is the scarcity of labeled, real-world datasets. Traditional models often fail under zero-day attacks due to data limitations. To address this [17, 18], proposed Conditional Tabular Generative Adversarial Networks (CTGAN) to synthetically generate realistic traffic data and enhance model training through a discriminator-based framework. Similarly [19], introduced a lightweight deep learning model using convolutional layers and global average pooling to detect five botnet attacks, offering scalability and low resource requirements.

To improve detection accuracy in IoT networks [20], proposed an optimization-based approach using the Dung Beetle Optimizer (DBO) to tune CatBoost parameters. Federated learning has also been integrated into IDS and NIDS to ensure privacy and fault tolerance in fog architectures [21]. For access control [22], introduced a lightweight group key management model based

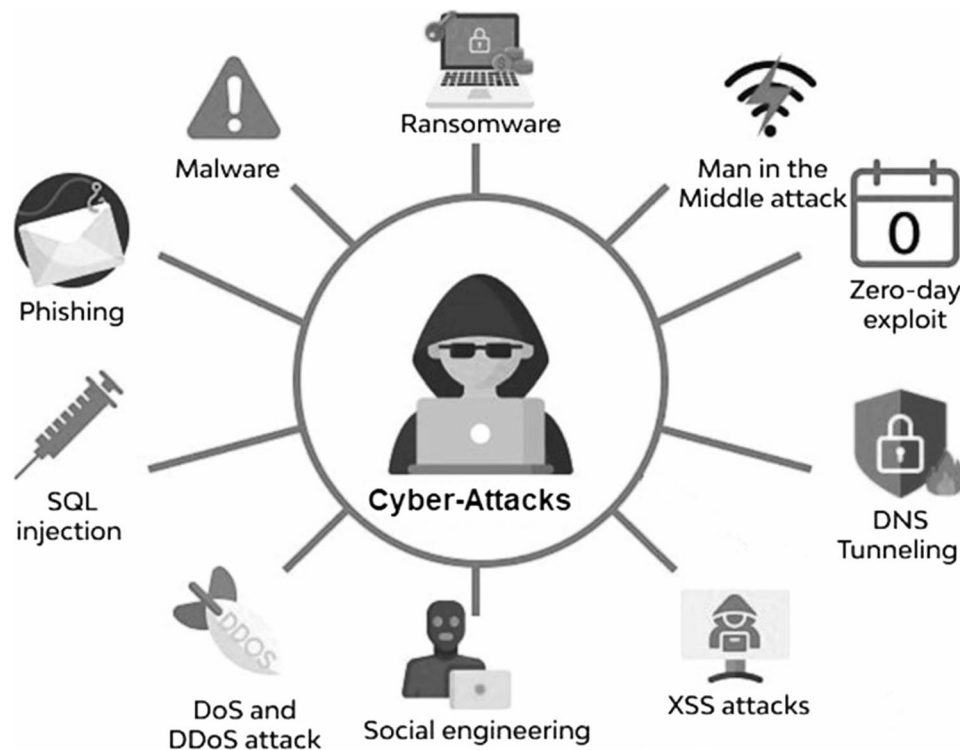


Fig. 1 Several frequent threats or attacks in the realm of cyber-security [8]

on Hyperledger Fabric, reducing resource consumption and enhancing secure communication in IoT. On the cryptographic side [23], developed a multi-dimensional chaotic encryption scheme for secure, low-power data transmission in resource-constrained environments. Firmware-level vulnerabilities have also been identified as critical weak points in smart home devices, with [24] recommending reverse engineering, standardization, and hybrid frameworks for robust protection.

Device-specific identification techniques have also evolved. For example [14], proposed enhanced device identification to secure component-to-component communication. To improve protocol-level security [25], evaluated ML classifiers (DT, KNN, RF, AdaBoost, XGBoost) for anomaly detection in MQTT systems, with RF performing best. Further comparative studies like [26] examined six machine learning models and three feature extraction methods across various datasets. Principal Component Analysis (PCA) and Adversarial Autoencoders (AE) outperformed Linear Discriminant Analysis (LDA) under dimensionality constraints. These findings emphasize the need for standardized benchmark features in NIDS development. Recent work also includes a cloud-optimized NIDS using transformer models with attention mechanisms for improved adaptability [27], as well as CNN-LSTM-based models integrated with SDN systems for real-time DDoS mitigation through anomaly detection and traffic flow control [28]. A novel knowledge

graph-based detection model using FT-CNN-BiLSTM-CRF architecture demonstrated faster and more accurate multi-step attack detection [29, 30].

The contributions of the study are:

- The main objective of this work is to evaluate the identification of cybersecurity risks including benign activity, DDOS-PSH-ACK and DDOS-ACK by machine learning algorithms.
- The paper also addresses issues on the implementation of machine learning (ML) security measures.
- This work evaluates the proposed and current models using metrics including accuracy, error rate, and detection rate.
- This paper examines the shortcomings of present models and proposed a reasonable solution to help improve understanding and prediction of cyberattacks.
- Furthermore investigated in this work are the several purposes of machine learning (ML), deep learning (DL), and the Internet of Things (IoT) in cyberattacks.

This study proposes a hybrid machine learning model that integrates Fuzzy C-Means clustering with Support Vector Machines (SVM) to enhance the detection of cyberattacks, specifically focusing on distinguishing

between DDOS-PSH-ACK, DDOS-ACK, and benign traffic. The proposed approach is evaluated alongside other conventional classifiers such as AdaBoost, Multi-Layer Perceptron (MLP), and standalone SVM. To provide a comprehensive assessment of model performance, we employed multiple evaluation metrics including accuracy, precision, error rate, detection rate, and the Matthews Correlation Coefficient (MCC), which offers a more balanced measure, especially under class imbalance. The study addresses key challenges in cybersecurity detection, such as overlapping classes, limited labeled data, and the complexity of real-world threats. Furthermore, the research explores the practical roles and limitations of machine learning, deep learning, and IoT in the broader context of cybersecurity. The findings demonstrate that the proposed hybrid model significantly outperforms traditional methods in both predictive accuracy and robustness, providing a practical and scalable solution for cyberattack detection systems.

The significance of this study lies in its targeted approach to detecting Distributed Denial of Service (DDoS) attacks specifically DDOS-PSH-ACK and DDOS-ACK using a hybrid machine learning model that combines Fuzzy C-Means clustering with Support Vector Machines (SVM). Unlike previous research that often generalizes detection over mixed attack types or applies standalone classifiers, our work focuses on enhancing detection accuracy in imbalanced datasets, a common challenge in real-world network environments. By integrating clustering to pre-process noisy data and improve class separability, the proposed model addresses overlapping class boundaries that hinder traditional models. Additionally, we provide a comprehensive performance evaluation using not only accuracy but also precision, detection rate, error rate, and the Matthews Correlation Coefficient (MCC), which offers a balanced assessment even with skewed data. This combination of methodological refinement and metric diversity contributes to a more robust and scalable DDoS detection system, filling existing gaps in the literature where many studies fall short in handling imbalanced or attack-specific data scenarios.

The following is how the paper is divided into sections: The introduction is covered in part 1, and Sect. 2 offers a literature analysis with specifics on cyberattack prediction methods. The suggested technique and the results are explained in detail in Sects. 3 and 4, and the paper's conclusion is given in Sect. 5.

Literature review

The growing complexity of modern networks has made intrusion detection a critical component of cybersecurity research. Several studies have explored the application of machine learning (ML) and knowledge discovery techniques to detect malicious activities in conventional

and IoT-based systems. Early works highlighted the utility of expert systems in recognizing anomalies and malicious behaviors. Recent advancements have enabled more accurate identification of cyberattacks through techniques like anomaly detection, pattern recognition, and behavioral analysis. Machine learning algorithms such as Random Forest (RF), Neural Networks (NN), and Autoencoders have demonstrated high effectiveness in intrusion detection. Particularly, unsupervised learning techniques are preferred for identifying zero-day attacks due to their ability to generalize patterns [30]. For instance [31], used autoencoders for feature extraction to streamline breach detection. These models, trained on normal and abnormal traffic, can effectively identify deviations in network behavior [32].

Deep learning approaches like Convolutional Neural Networks (CNNs) further improve intrusion detection. Each CNN layer identifies correlations within input features, with SoftMax classifiers providing high-accuracy predictions [33]. A deep neural network-based system was also developed to detect botnet activities, where honeypots captured malicious traffic, converted it into feature vectors, and classified it as benign or malicious [34]. Feature selection remains a critical step in optimizing model performance. In [35], four ML algorithms K-Nearest Neighbors (KNN), RF, Support Vector Machine (SVM), and Decision Tree (DT) were evaluated. RF outperformed others, achieving a accuracy of 99.72%. Similarly [36], developed a SHAP (SHapley Additive exPlanations)-based model to interpret classifier behavior, enhancing transparency and performance.

Intrusion detection in Industrial Control Systems (ICS) and IoT environments is particularly challenging due to their complexity and interconnectivity. In [37], a hybrid Graph Convolutional Network (GCN) and Long Short-Term Memory (LSTM) model was proposed to capture dynamic interactions and improve threat detection in these systems. To strengthen IDS [38, 39], emphasize the need for improved data flow mechanisms, such as vector networks, and discuss how deep learning models have evolved to handle modern attacks. Malware and spam detection have also been enhanced using ML models that integrate attention mechanisms to improve precision [40].

Several surveys have summarized the effectiveness of ML methods like Deep Learning, Ensemble Learning, Naive Bayes, K-Means, and Artificial Neural Networks for IDS. These reviews discuss the role of classifiers like SVM, Decision Trees, and KNN in enhancing detection accuracy and system performance. A notable distinction in [41, 42] is its focus on evaluating ML algorithms on DDoS-specific datasets, in contrast to prior studies that used broader datasets. This targeted approach provides critical insights into classifying DDoS traffic, contributing

significantly to real-world mitigation strategies. Hybrid models have become increasingly prominent in improving detection accuracy and reducing training time.

A hybrid ML model presented in [43] combined learning models in parallel and sequential phases but achieved only 86.1% accuracy, which is suboptimal for real-world deployment. The model in [44] improved upon this by combining clustering with SVM, where data was first clustered and then used to train separate models, increasing recognition rates. Further hybrid methods have achieved higher detection rates. The authors in [45–49] proposed enhanced hybrid models that reached accuracy levels of 87.0%, 98.62%, and 99%, respectively. These results underscore the importance of hybrid machine learning techniques in advancing Cyberattack Detection Systems (CADS) and their continuous optimization for better performance.

Several domainspecific hybrid ML frameworks have been proposed for largescale DDoS mitigation. BotDefender, a collaborative defense model using network traffic analysis and machine learning, targets botnetbased DDoS attacks by combining anomaly detection and classifier techniques to proactively isolate malicious traffic. Another notable framework specifically addresses reflectionbased and amplified DDoS threats using a feature selection method (Closeness Index of Features) coupled with ensemble voting classifiers. Evaluated on the CICDDoS2019 dataset, this model achieved exceptionally high performance (99.99% accuracy, with strong MCC values), demonstrating resilience to reflection-based volumetric attacks.

By positioning our work alongside these frameworks, we highlight the value of hybrid approaches and justify our choice to integrate Fuzzy CMeans clustering and SVM a combination that similarly targets class overlap and imbalance, especially in highvolume attack scenarios.f (Table 1).

This study contributes to the cybersecurity domain by demonstrating the impact of machine learning (ML) techniques specifically on Distributed Denial of Service (DDoS) attack detection. Unlike broader analyses that cover a mix of attack types, this research provides an in-depth assessment of algorithmic performance on DDoS-focused datasets, addressing a significant gap in the existing literature. To enhance DDoS detection in fog-IoT systems, the authors in [53] proposed the ReputE framework, which utilizes reputation-based trust modeling and a soft-voting ensemble of classifiers. The framework targets common DDoS and DoS attacks using an active search-based feature selection mechanism. Its evaluation across multiple benchmark datasets including NSL-KDD, CICDDoS2019, IoTID20, N-BaIoT2018, TON_IoT, and UNSW_NB15 demonstrated the framework's adaptability and robustness. Similarly [54], introduced P2ADE,

a privacy-preserving detection framework based on stacked ensemble learning. This system effectively identifies man-in-the-middle (MiTM) and DDoS attacks in IoT environments with an accuracy of up to 99.98%. It employs dimensionality reduction through Feature Dimensionality Management (FDM), which reduces dataset size by over 97% while maintaining 99.9% detection accuracy. The improved variant, DFDM, further enhances attack identification performance.

In healthcare IoT [55], presented *Holo-BlockChain*, a hybrid architecture that combines Holochain and Blockchain technologies. This framework provides secure, low-overhead data management, outperforming conventional blockchain systems in terms of computational and storage efficiency. The author of [56] introduced a CNN-based transfer learning model designed for 5G IoT scenarios. Tested on BoT-IoT and UNSW-NB15 datasets, it achieved 97.9% accuracy and showed high reliability in detecting zero-day attacks. Meanwhile [57], developed a hybrid IDS combining Recurrent Neural Network (RNN) and Gated Recurrent Units (GRU), which achieved nearly 99% accuracy on the ToN-IoT dataset across multiple IoT communication layers. Emerging trends show a convergence of blockchain and federated learning techniques in IoT security. A recent survey [58] emphasised this integration, identifying research directions such as decentralised trust models and privacy-preserving data sharing. One such framework, *FedBlockHealth*, applies federated learning and blockchain to securely manage and share sensitive IoT healthcare data while ensuring user privacy [55].

Dataset description and exploratory analysis

This study utilizes the Cyber-Attacks Detection (APA-DDoS) dataset, which is publicly available on Kaggle [h <https://www.kaggle.com/datasets/yashwanthkumbam/apaddos-dataset>]. The dataset comprises labeled network traffic data specifically designed for evaluating DDoS detection systems. It includes various classes of network behavior, such as benign traffic, DDOS-PSH-ACK, and DDOS-ACK attack instances. These features are selected to ensure a focused and realistic evaluation of machine learning-based intrusion detection systems in identifying targeted DDoS threats. The APA-DDoS dataset contains 24 features, including both numerical and categorical values such as protocol, timestamp, packet_length, flags, and more. Each instance is labeled to facilitate supervised learning methods. One of the key strengths of this dataset is its clear separation of attack types, which supports multi-class classification tasks and helps in evaluating the detection capability of various models under imbalanced data distributions. Exploratory Data Analysis (EDA) was performed to understand the structure and distribution of the data. Initial findings show that the dataset is

Table 1 Existing literature table

Study	Focus Area	Methodology	Key Findings	Limitations/Gaps
[30]	Botnet detection in IoT	Machine learning techniques for network forensics	Proposed ML-based detection for botnet activities	Limited dataset, lacks real-time evaluation
[31]	Intrusion detection	Autoencoders for feature extraction	Autoencoders improve intrusion detection accuracy	High computational cost, requires extensive training
[32]	DDoS attack identification	Deep learning for network traffic analysis	Deep defense model effectively identifies DDoS attacks	Limited to known attack patterns, struggles with novel threats
[33]	Botnet detection	Deep learning on network flow summary	Used SoftMax classifier for identifying malicious traffic	Does not address encrypted traffic, lacks real-world validation
[34]	Botnet detection	Honeypot-based system for malware analysis	Improved honeypot system for intentional infection	Cannot detect botnets spreading via social networks
[38]	Intrusion detection	Deep learning-based IDS	Demonstrated deep learning's effectiveness in IDS	High false positive rate, lacks interpretability
[47]	Intrusion detection	ML-based comparative analysis	Compared ML techniques for IDS improvement	Does not address adversarial ML threats
[41]	Distributed IDS	Two-phase IDS with ML	Improved IDS efficiency using Hadoop and ML	Scalability issues, lacks real-time processing
[43]	Intrusion detection	Neural architecture search	Enhanced intrusion detection efficiency	High model complexity, computational overhead
[48]	Cybersecurity language	Rhetorical analysis	Examined cybersecurity communication strategies	Lacks technical depth, focuses only on language aspects
[42]	Hybrid IDS	Systematic review	Proposed hybrid IDS for better threat detection	No empirical validation, relies on theoretical analysis
[44]	Cyberattack detection	Boosted ensemble ML	Developed an efficient ML-based intrusion detection model	High memory usage, limited scalability
[45]	Cyber-physical security	Hybrid deep learning attack graph model	Achieved 87% accuracy in cyber-physical security	Limited to specific power systems, lacks generalization
[46]	Industrial control system security	Graph-aware deep learning model	Achieved 98.62% accuracy in attack detection	High dependency on labeled data, difficult to scale
[49]	DDoS detection	Time-series based ML models	Reached 99% accuracy in DDoS attack detection	Only evaluates time-series data, limited model generalization
[50]	Cyberattacks & threats	IoT, Cyberattacks and threats based on ML and DL models	Compared ML and DL techniques for Cyberattacks and threats improvement	Only compared ML and DL techniques for Cyberattacks and threats improvement
[51]	Wireless sensor networks security	WSN security based on different ML models	Examined WSN security and ML algorithms	Does not address WSN security, intrusion and error detection
[52]	Cybersecurity & Automated distribution network	Cybersecurity and different types of attacks on the smart grid based on ML and DL models	Compared different types of attacks that smart grids and Cybersecurity	Only compared ML and DL techniques for Cybersecurity and smart grids improvement

imbalanced, with benign traffic comprising a significant portion compared to attack classes. Therefore, preprocessing steps such as normalization, label encoding, and resampling were carried out to improve model performance. Correlation heatmaps were also used to identify strongly related features, and redundant or highly correlated attributes were removed to reduce dimensionality and overfitting risk.

A study publication [59] says that AttackNet is a powerful deep learning model for finding and sorting different types of botnet attacks in IIoT. This proposal is based on the adaptable CNN-GRU architecture. The model protects IIoT networks from advanced cyberattacks with a testing accuracy of 99.75%, a precision score of 99.75%, a recall score of 99.74%, and a loss of 0.0063. We provide

a full study of the model using the most recent dataset and well-known performance evaluation measures.

The author used a publicly available dataset that had both hazardous and regular network traffic for their study [60]. The packets came from a real-time lab with cameras that was connected to the Internet of Things. The evaluation measures show that the proposed approach works well. The model that was suggested gets 99.52% of the tests right, 99.55% of the training right, and 0.0150% of the losses.

Cyber-Sentinet is a deep learning-based intrusion detection system (IDS) that is made to secure cyber-physical systems in industrial Internet of Things (IIoT) scenarios. This paper [61] suggests Cyber-Sentinet as a way to fix these concerns. Cyber-Sentinet is different from typical intrusion detection systems (IDS) because

it uses Shapley Additive Explanations (SHAP) to make its decision-making process clearer to grasp. This gives security experts more faith in the system's detections and helps them comprehend them better. Cyber-Sentinet has been shown to work well after a lot of testing on the Edge-IIoT-2022 dataset, which includes a variety of cyber-attacks such distributed denial of service, SQL injection, and man-in-the-middle assaults.

The study asks four research questions about the security of the Internet of Things (IoT) using machine learning and deep learning methods. To reach the goal, this is done. It also gives a full review of the most recent publications and comes to the conclusion that the security problems with the Internet of Things (IoT) can be fixed by using a mix of specialized solutions. According to the study's conclusions [62], a thorough investigation of the literature could greatly improve the security of Internet of Things (IoT) devices and apps, protecting user data from unwanted access and probable misuse.

Proposed methodology

With the increasing sophistication of cyber threats, traditional security mechanisms are often insufficient in detecting and mitigating attacks. Machine learning-based cybersecurity solutions provide a more adaptive and intelligent approach to threat detection. The Cyber-Attack Detection System (CADS) presented in this study leverages advanced machine learning techniques to identify and classify malicious activities with high accuracy. Unlike conventional methods, this system integrates fuzzy C-Means clustering with classification models to enhance detection efficiency.

The key differentiator of this study lies in the combination of SVM, AdaBoost, and MLP with fuzzy C-Means clustering, which optimizes accuracy, precision, recall, sensitivity, and specificity. This structured approach ensures a more robust detection system capable of handling large-scale cyber threats effectively. The methodology follows these steps:

1. Data Collection: The dataset is retrieved from Google Drive and loaded into the system.
2. Data Preprocessing: The data undergoes multiple preprocessing steps, including data cleaning, feature selection using the random forest algorithm, standardization via standard scaler, and normalization to enhance model performance.
3. Standardization: A structured system is developed to transform frequencies and amplitudes, ensuring consistent output analysis.
4. Clustering: Fuzzy C-Means clustering is applied to improve classification metrics such as accuracy, precision, recall, F1-score, sensitivity, and specificity.
5. Data Splitting: The dataset is divided into 75% training data and 25% testing data for model evaluation.
6. Model Development: Machine learning classifiers, including Support Vector Machines (SVM), AdaBoost (ADB), and Multi-Layer Perceptron (MLP), are implemented to construct the final detection model.

By integrating machine learning with clustering techniques, this study proposes a more adaptive and efficient cybersecurity detection system compared to conventional methods. The findings demonstrate that this approach significantly improves detection accuracy, making it a valuable contribution to cybersecurity research.

Figure 2, which provides a summary of the study project and its breakdown structure, graphically depicts the design of a cyber attack detecting system.

The process of retrieving data from a file is depicted in the flow chart, and data pretreatment is carried out by standardizing and normalizing the data using data-cleaning procedures. The training and testing data are processed using classification and clustering algorithms to perform modal validation. The algorithms are separated into two sections, classification and data preprocessing, each of which has a distinct function throughout the entire process.

Preprocessing

Cleaning and clustering procedures are used in the preprocessing stage of data to retrieve relevant information. Applications of fuzzy c-means are explained below. The classification stage will involve additional processing and manipulation of the data.

Data collection

Cyber attack detection system (CADS) datasets were obtained using the Kaggle platform, which offers a variety of machine learning datasets for study. This dataset consists of internet traffic logs that were monitored by the Cyber-Attacks intrusion detection network. They show traffic traces found by an actual CADS, with only fragments of evidence remaining to verify its existence. There are 11 network-related attributes and 151,200 tuples in this data collection. The target class denotes each distinct outcome state. There are 37,800 cases classified as DDoS-PSH-ACK, 75,600 as Benign, and 37,800 as DDoS-ACK. A comparison analysis of the frequency of cyberattacks among DDoS-ACK, DDoS-PSH-ACK, and Benign traffic is shown in Fig. 3. The DDoS-ACK and DDoS-PSH-ACK bars represent cyberattacks, while Benign traffic is represented separately. This figure provides important insights

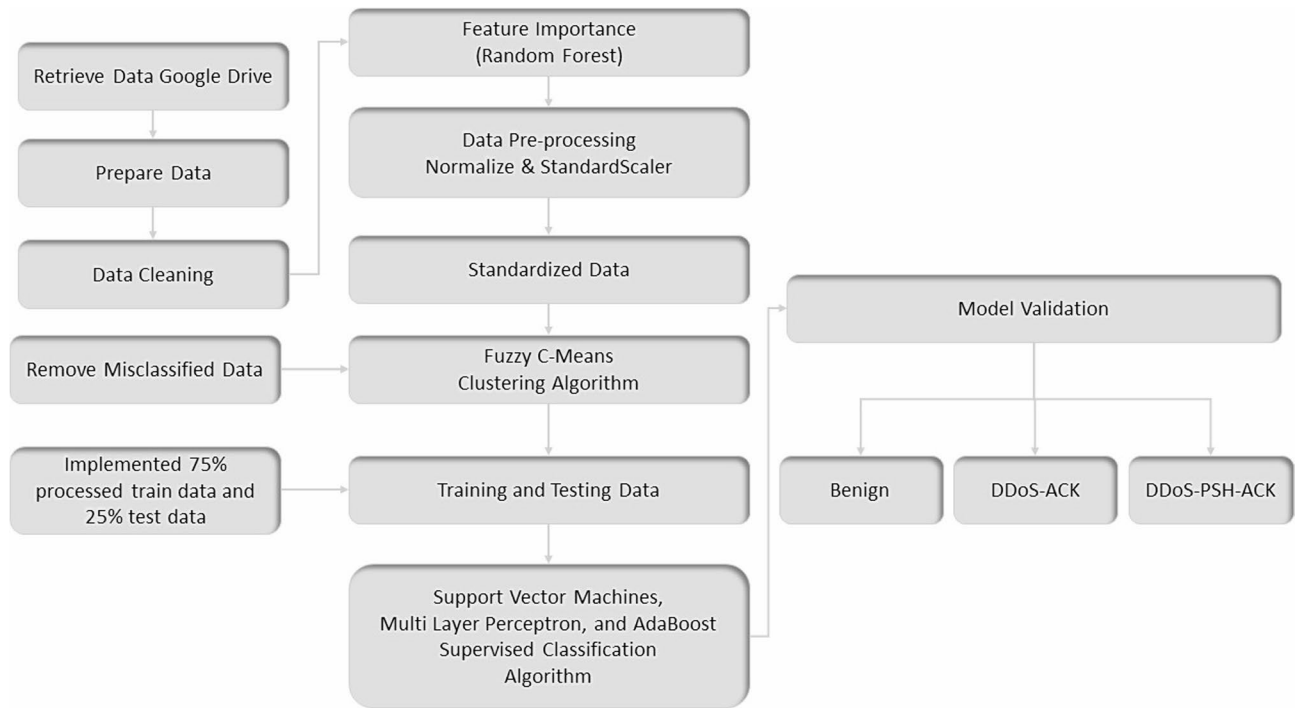


Fig. 2 Cyber-Attacks Detection System (the CADS) Proposed Method

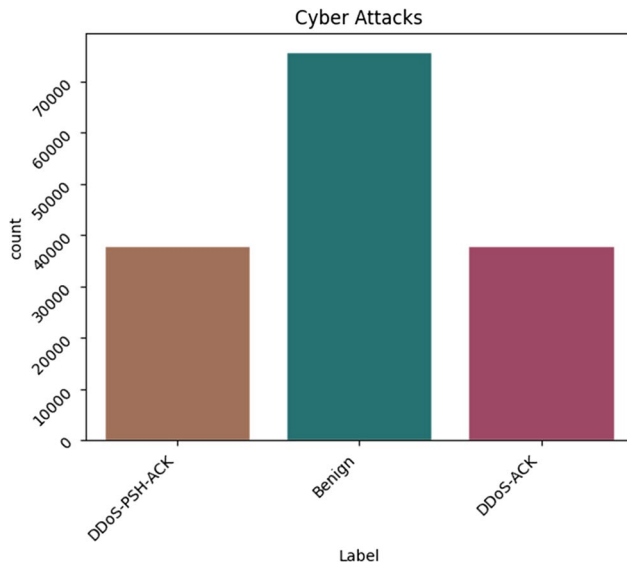


Fig. 3 Facts of Cyber-Attacks numbers

into cyberattacks and highlights discrepancies in cyber threat distribution.

Feature selection and recursive feature elimination

One method for selecting the best features in a dataset is recursive feature elimination. It entails eliminating the least significant characteristics from an iteration based on their relevance to the objective variable, as determined by machine learning techniques such as XGBoost, random forest, and support vector machine. Conventional

RFE methods depended on the prediction's performance, and pertinent features were evaluated using mathematical and strategic processes. Better feature selection outcomes and increased computing efficiency are achieved by utilizing the capabilities of many algorithms to identify and eliminate the least useful qualities.

Recursive Feature Elimination (RFE) was selected due to its strong integration with the underlying model, particularly Support Vector Machines (SVM), which ranks features based on their importance in the classification task. While alternatives such as Chi-Square or LASSO are faster, RFE provides more precise selection in cases where feature interdependence and non-linearity affect model performance common traits in cybersecurity data. Preliminary testing showed that RFE consistently improved model accuracy and robustness compared to simpler filter methods, justifying its computational cost.

A concrete example will be given below to demonstrate the RFE choice of attributes approach:

RecursiveFeature

$$In = \frac{E(y)}{1 - E(y)} = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} \text{Elimination} \quad (1)$$

The cyber attack detection system's key features are scored in depth in Table 2, which also offers pertinent insights. A graphical depiction of key characteristics that aid in cyberattack prediction models is shown in Fig. 2.

Table 2 Cyber-Attacks detection system (CADS) feature importance dataset

Sr. No.	Feature	Feature Importance Score
1	tcp.flags.push	0.382107
2	Tx Bytes	0.117767
3	TCPsrcport	0.110976
4	frame.Len	0.110716
5	ip.flags.df	0.096411
6	Rx Bytes	0.072806
7	Bytes	0.055349
8	Tx Packets	0.032502
9	Packets	0.015106
10	Rx Packets	0.006259

The names and scores of the main features for CADS are described in Table 2, and a visual representation of the key features is displayed in Fig. 4.

Figure 4 illustrates the feature importance analysis of the Cyber-Attacks Detection System (CADS) using the APA-DDoS dataset, highlighting which network attributes most significantly influence the detection of cyber threats. The bar and line charts reveal that tcp.flags.push is the most dominant feature, followed by Tx_Bytes, tcp.srcport, and frame.len, indicating that TCP flag behaviors and data transmission characteristics are critical for distinguishing between benign and malicious traffic. In contrast, features like ip.proto, tcp.ack, and tcp.seq contribute minimally to the model's predictive power. The steep decline in the line chart demonstrates that a small subset of features carries the majority of the model's discriminative capability, justifying the use of feature selection techniques like Recursive Feature Elimination (RFE) to improve model efficiency and reduce computational overhead. This analysis supports the effectiveness of the proposed CADS model in accurately identifying DDoS attacks while optimizing performance.

The current study investigates the role of crucial processes in predicting cyberattacks. The steps include data normalization, simulation, pre-processing, and

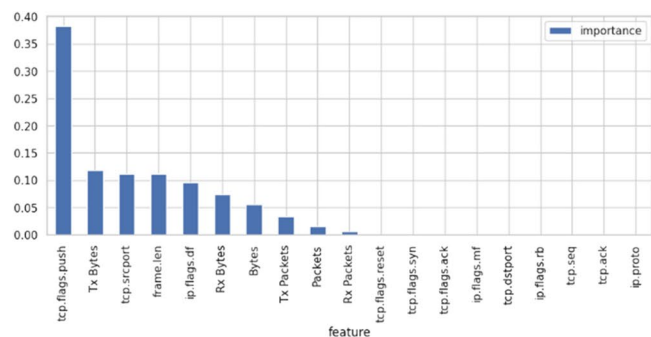
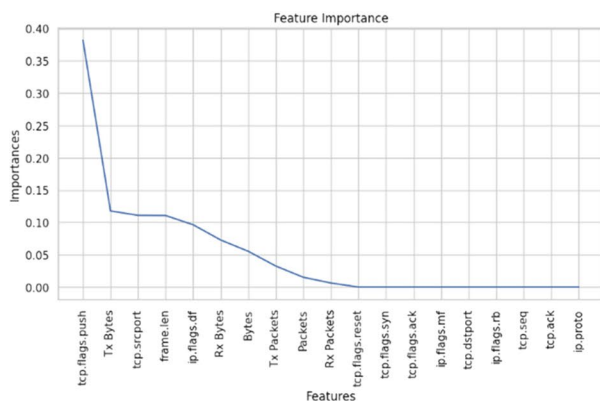
Table 3 Shows the dataset for predicting cyber-attacks, which has been processed

array ([[−2.25600828, 3.17966878, 4.07155662, ..., 2.79612371, 2.8913428, 3.79623039],
[−3.63211306, 3.12297205, 4.01149194, ..., 3.63819285, 3.82059211, 3.73816997]
[−5.24328435, 3.05694891, 3.94154682, ..., 4.44515977, 4.71110505, 3.6705588], ...,
[0.28229919, −0.81551329, −0.53269127, ..., −0.5691574, −0.37483052, −0.65438172],
[0.27796174, −0.81554572, −0.53269127, ..., −0.5691794, −0.3372011, −0.65438172],
[0.28204024, −0.81555166, −0.53269127, ..., −0.56918343, −0.37247629, −0.65438172]]])

induction. Critical criteria, post-processing, complexity issues, and system efficacy are also investigated. The process of pre-processing and data preparation, including standardization and normalization, begins with the collection of data from the dataset. The final pre-processed and cleaned information is shown in Table 3.

Table 3 displays the normalized and preprocessed dataset used for cyber-attack prediction, where each row represents a network traffic instance and each column corresponds to a scaled feature. The transformation ensures all feature values are within a similar range, improving model performance and training stability. Higher values in some rows may indicate attack patterns, while lower or negative values suggest normal traffic. This standardized format is essential for accurate and efficient machine learning-based detection of cyber threats like DDoS attacks.

Figure 5 shows the mixed information Chat cyberattack detection system. Figure 5 shows a scatter plot of mixed data used in the Cyber-Attacks Detection System (CADS). The data forms curved, radial patterns, indicating complex, nonlinear relationships between features. The dense cluster near the origin suggests a concentration of similar data points, helping to visually separate normal from malicious behavior. This supports the need for advanced models to detect diverse cyber-attack

**Fig. 4** Cyber-Attacks Detection System (CADS) Feature Importance Dataset

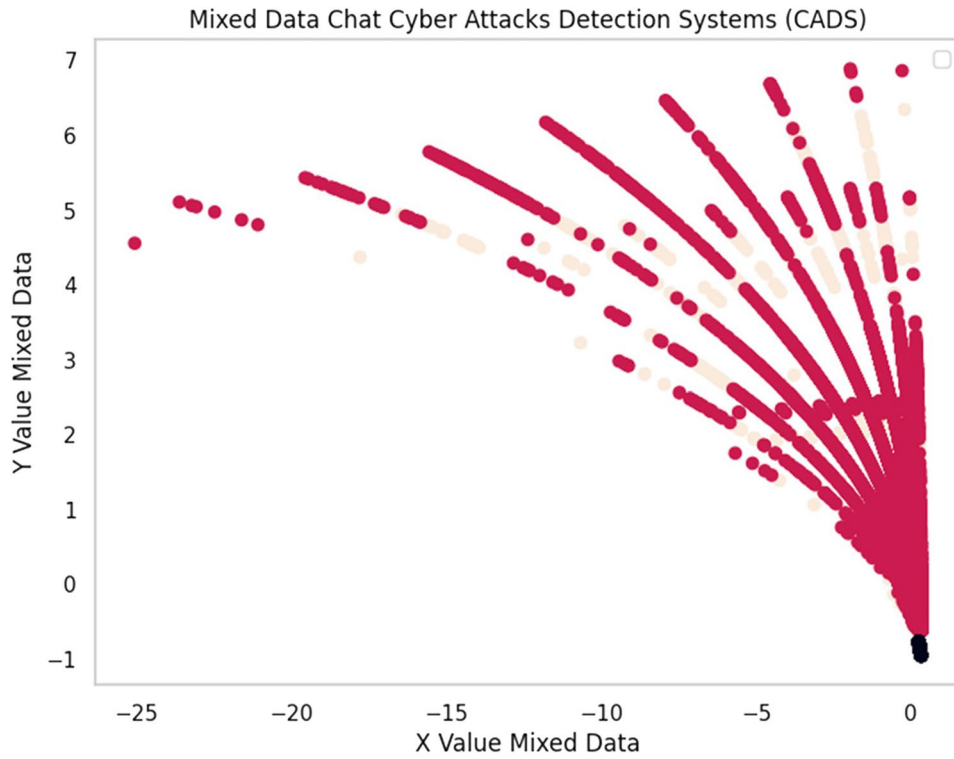


Fig. 5 Mixed information chat cyber attacks detection system

patterns. Furthermore, sensitive data is graphically depicted requiring fuzzy C mean implementation. The Y value represents the yellow color circle, and the X value represents the purple colour circle.

Fuzzy C-means clustering method

Clustering is a prominent form of unsupervised learning with a wide range of applications in numerous sectors. Clusters are formed by organizing and analyzing data, and each cluster is given a unique assignment. The most widely used clustering approach is fuzzy-C means. The FCM approach assigns a membership value to each data point depending on its distance away from the centroid cluster. When the information point is near the centroid, its membership score will be higher. There should be a single, clear membership value for each data piece. After each iteration, the cluster centroid is updated using the formula below:

$$\mu = 1 / \sum_{k=1}^c (d_{ij} / d_{ik})^{(2/m-1)} \quad (2)$$

$$v_j = \left(\sum_{i=1}^n (\mu_{ij})^m x_i \right) / \left(\sum_{i=1}^n (\mu_{ij})^m \right), \forall j = 1, 2, 3, \dots, c \quad (3)$$

Where,

The number of data points is denoted by 'n'	'vj' shows the jth cluster centre
'm' shows fluffiness directory $m \in [1, \infty]$	'c' shown the number of cluster centres
the jth cluster centre and ith data link are denoted by 'uij'	the jth cluster centre and ith data are indicated by 'did' Euclidean distance

The main objective of the c-means fuzzy method is to reduce:

$$J(U, V) = \sum_{i=1}^n \sum_{j=1}^c (\mu_{ij})^m \|x_i - v_j\|^2 \quad (4)$$

' $\|x_i - v_j\|$ ' symbolizes the cluster center of the Euclidean jth and ith points of data. Different functions or techniques for mathematical separation can be used by different algorithms. Based on category, the most widely used methods are Manhattan, Euclidean, and Hamming distances. Among these methods are:

$$\text{Euclidean} \quad \sqrt{\sum_{i=1}^k (x_i - y_i)^2} \quad (5)$$

$$\text{Manhattan} \quad \sum_{i=1}^k |x_i - y_i| \quad (6)$$

$$\text{Minkowski} \left(\sum_{i=1}^k (|x_i - y_i|)^q \right)^{1/q} \quad (7)$$

The steps of Fuzzy C-Means clustering:

Let $X = \{x_1, x_2, x_3, \dots, x_n\}$ be the representation of a set of data points, and $V = \{v_1, v_2, v_3, \dots, v_c\}$ is the representation of a set of centres.

1. 'c' cluster centroids are selected randomly.
2. ' μ_{ij} ' is the fuzzy participation and can be calculated by this formula.

$$\mu_{ij} = 1 / \sum_{k=1}^c (d_j / d_{ik})^{2/m-1} \quad (8)$$

3. Computation of fuzzy centroids ' v_j ' as follows

$$v_j = \left(\sum_{i=1}^n (\mu_{ij})^m X_i \right) / \left(\sum_{i=1}^n (\mu_{ij})^m \right), \forall j = 1, 2, 3, \dots, c \quad (9)$$

4. Until the achievement of $||U(k+1) - U(k)|| < \beta$ or the smallest 'J' value keep repeating steps 2 and 3

- I) the letter 'k' indicates the repetitive steps
- II) ' β ' defines the completion criteria between [0, 1]
- III) ' $U = (ij) n \times c$ '. defines the matrix of fuzzy relationship
- IV) the letter 'J' represents an impartial function

Preprocessing techniques are used to create mixed data representations from standard datasets. Fuzzy C-Means applications enable the processing and filtering of large data sets by eliminating redundant information. By clustering and giving each an option or probability for deciding cluster membership, three distinct clusters are found. This method creates a membership matrix that shows the relationship between each sample and the cluster. Values for centroid clustering The methodology for a variable dataset with multi-class data uses the fuzzy c means algorithm. Every data point has a centroid, and clusters with closer proximity to the data set have stronger relationships. This CADS data collection contains eleven dimensions, including a target characteristic that shows cluster counts and cyberattack-related properties. The data is visually shown using graphs of lines and clusters

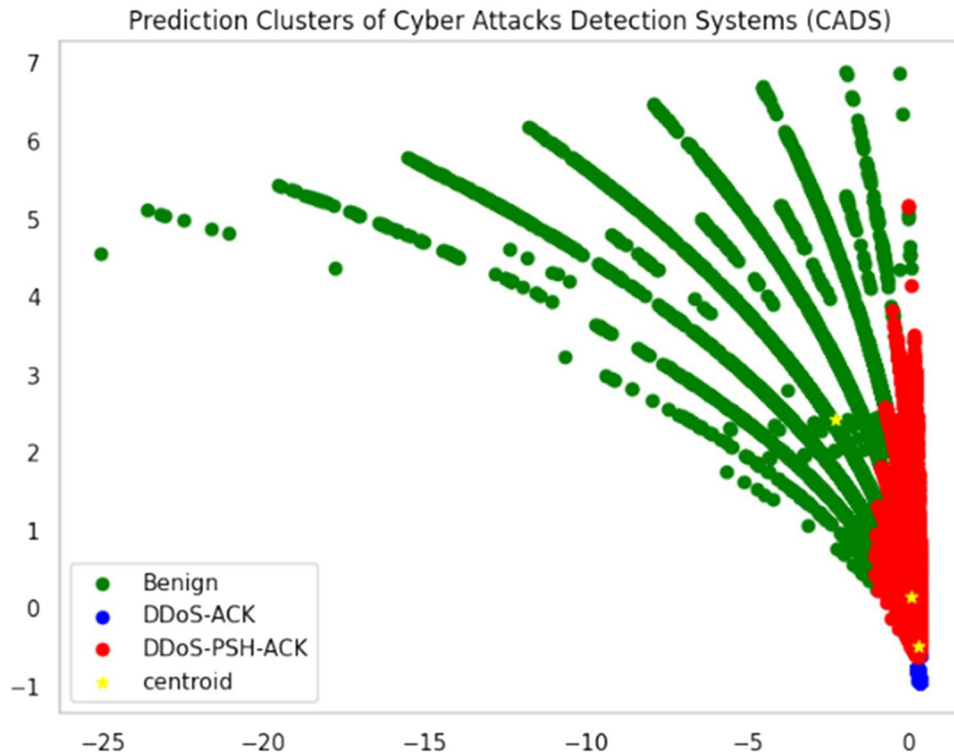


Fig. 6 Fuzzy C-means three clusters (HDP)

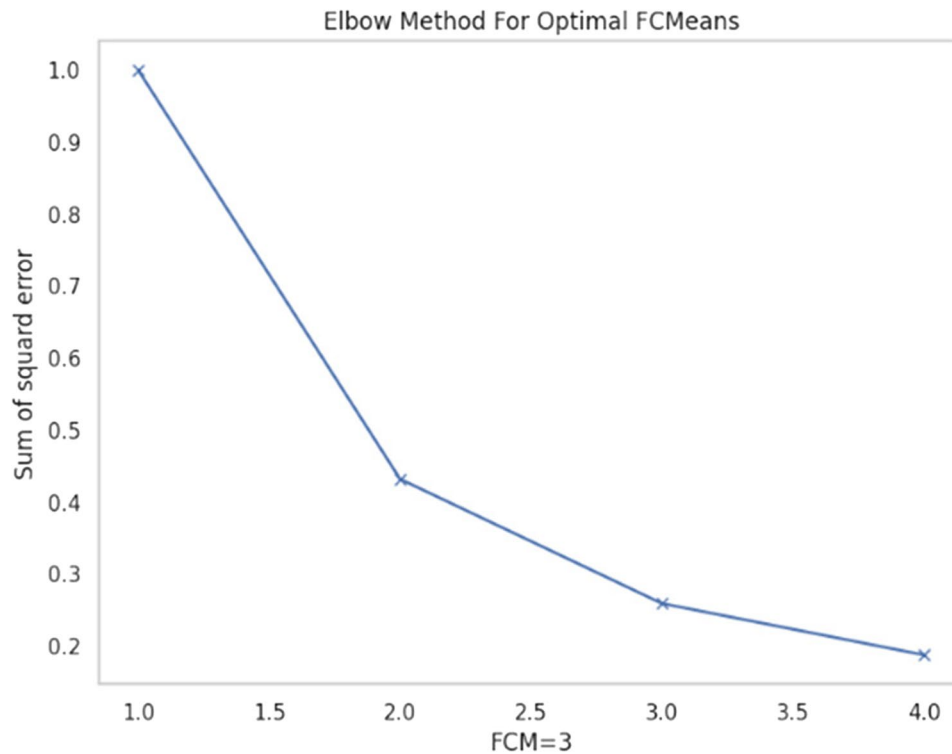


Fig. 7 Fuzzy C-Means Sum of Squared Error Line Chart

Table 4 Value of the fuzzy C-Means classification centroid

```
array ([[[-2.26290809, 2.42248155, 1.94869573, -0.97126085,
2.63626742, 2.69777945, 2.64506986, 2.6906121, 2.6249295, 2.70252328,
2.38238635],
[0.29153648, -0.49108827, -0.38221559, 0.84068189, -0.56276646,
-0.47419687, -0.54924381, -0.48091388, -0.57539634, -0.46693047,
-0.60568271],
[0.0686699, 0.14751575, 0.16005833, -0.96365286, 0.26531037,
0.1324408, 0.24514482, 0.14301613, 0.28474029, 0.12157745,
0.35106418]])
```

Table 5 Three clusters of fuzzy C-Means cyber attacks detection system (CADS) Pre-processed predictive dataset

```
array ([[[-2.25600828, 3.17966878, 4.07155662, ..., 2.8913428,
3.79623039, 1.], [-3.63211306, 3.12297205, 4.01149194, ..., 3.82059211,
3.73816997, 1.,
[-5.24328435, 3.05694891, 3.94154682, ..., 4.71110505, 3.6705588,
1.], ..., [0.28229919, -0.81551329, -0.53269127, ..., -0.37483052,
-0.65438172, 2.], [0.27796174, -0.81554572, -0.53269127, ...,
-0.3372011, -0.65438172, 2.],
[0.28204024, -0.81555166, -0.53269127, ..., -0.37247629,
-0.65438172, 2.]])
```

in Fig. 6 and 7. Additionally, these images demonstrate how unstructured data is converted into structured data. Table 4 shows the value of the fuzzy C-means classification centroid and Table 5 shows the three clusters of fuzzy C-means cyber attacks detection system (CADS) pre-processed predictive dataset. Instead of the complete numerical arrays, the results are summarized using key statistical attributes. For example, Table 4 now reports

the mean, standard deviation, minimum, and maximum values for each metric across all experimental runs. Similarly, Table 5 presents aggregated statistical summaries to highlight performance consistency and variability.

The clustering feature detection is evaluated using a variety of clustering approaches, F-measure metrics, precision, and recall. System deviation is evaluated according to established criteria, and concern ratings are computed so that the result can be categorized as legitimate, suspicious, or unlawful. This method classifies data according to predetermined standards to determine the system's efficacy.

Classification

The process of classifying data involves using training data to divide data into distinct groups. The most effective classifier for the provided dataset is determined by supervised learning. Tests utilizing various classification algorithms will be used to assess the effectiveness and performance of various classifiers. In this manner, the best classifier for the particular dataset will be identified.

Method Of support vector machine

Support vector machines are used in supervised learning applications for regression and classification. This method divides the mixed classes in graphs into various groups within a maximum margin called high-dimension space. The SVM model uses mathematical systems such

as kernel functions, not linear, and linear functions like RBF, sigmoid, and polynomial to determine the best separation between two categories. In particular, data points of different classes are separated using decision boundary support vectors, which are made up of the two nearest points. By using mathematical regression and classification functions like Linear SVM, Non-linear SVM, and Kernel function, the SVM approach produces precise and efficient predictions.

$$\text{Linear SVM Model} \quad X_i.X_j \quad (10)$$

$$\text{Non - Linear SVM} \quad \phi(X_i).\phi(X_j) \quad (11)$$

$$\text{Kernel Function} \quad k(X_i.X_j) \quad (12)$$

There are several crucial stages in the SVM method:

- I. To maximize distances between classes and partition data, the method first chooses the optimal hyperplanes.
- II. Techniques to avoid misclassification are used while processing non-linear separable data.
- III. The technique improves decision surface recognition and separability to transform input data into a space with greater dimensions.

A popular statistic for tackling classification problems is the confusion matrix. Both binary and multiple classes can be used it. The Confusion matrix is typically examined to assess the CADs.

To provide a more balanced and informative evaluation of classification performance particularly in the presence of class imbalance the Matthews Correlation Coefficient (MCC) was computed. MCC considers all four confusion matrix categories (TP, TN, FP, FN) and provides a comprehensive performance score, ranging from -1 (total disagreement) to +1 (perfect prediction), with 0 indicating no better than random guessing.

The MCC is calculated as follows:

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)}} \quad (13)$$

- The correlation between correctly detected attacks and the total number of attacks is evaluated by the rate of true positives (TPR).
- When all intrusions are accurately detected, which is rare, a cyber-attack detection system (CADs) possesses a TPR of 1. The TPR is defined as the Detection Rate (DR), as it is more widely known.

$$\text{True Positive Rate (TPR)} = \frac{TP}{(TP + FN)} \quad (14)$$

- The no. of normal instances which are incorrectly classified as attacks relative to the overall amount of typical instances is known as the False Positive Rate. The following is how FPR is calculated:

$$\text{False Positive Rate (FPR)} = \frac{FP}{(FP + TN)} \quad (15)$$

- The intrusion detection system's failure to identify the intrusion and classify it as normal is shown by the False Negative Rate (FNR). The following is the FNR calculation.

$$\text{False Negative Rate (FNR)} = \frac{FN}{(FN + TP)} \quad (16)$$

The data is accurately projected into high-dimensional space throughout this process. The system can predict labels on both new and existing datasets after it has been trained. The objective is to generate predictions that accurately match the labels. The outcome of the confusion matrix, which shows how well the SVM method performs in terms of predictions, is shown in Fig. 8 and Fig. 9 [62].

In this research, the matrix of confusion is written as [[A B] [C D]], where:

- A stands for properly anticipated negative cases.
- B for incorrectly forecasted positive instances.
- C for incorrectly predicted negative instances.
- D for correctly predicted positive instances.

Multi-layer perceptron (Mlp) method

An MLP is another extensive optimization approach. It has a large number of perception neurons. They are made up of two separate levels: input, which receives data, and output, which assesses or estimates the input [63, 64]. The infinite hidden layer serves as the actual processing engine of MLP. With a variable number of hidden layers, MLP may approximate any continuous function. MLP and a hidden layer can be used to approximate any continuous function. The Multi-Layer Perceptron solves this issue by differentiating between sets that lack logical autonomy. To do this, participants construct machine prediction and learning methods for challenging data using increasingly intricate and adaptable frameworks. This method is frequently employed to solve supervised learning issues.

$$\text{Sigmoid} \quad S(z) = \frac{1}{(1 + e^z)} \quad (17)$$

$$\text{Regression} \quad y = e^{\wedge}(b_0 + b_1 * x) / (1 + e^{\wedge}(b_0 + b_1 * x)) \quad (18)$$

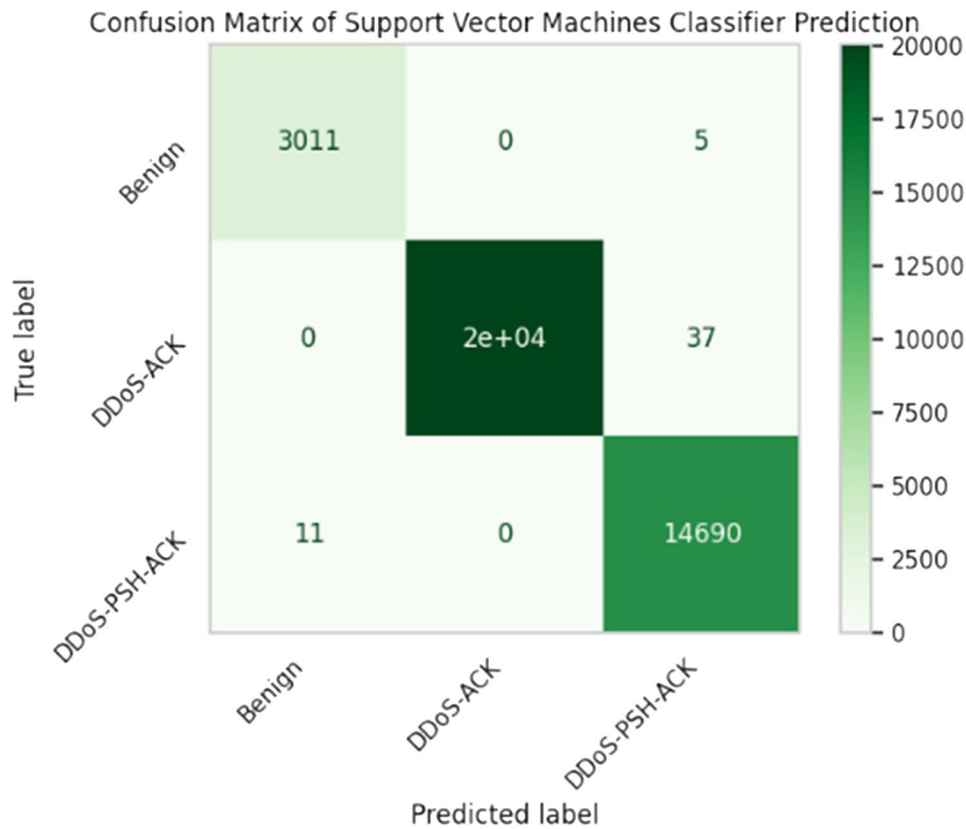


Fig. 8 Confusion Matrix Support Vector Machine Method

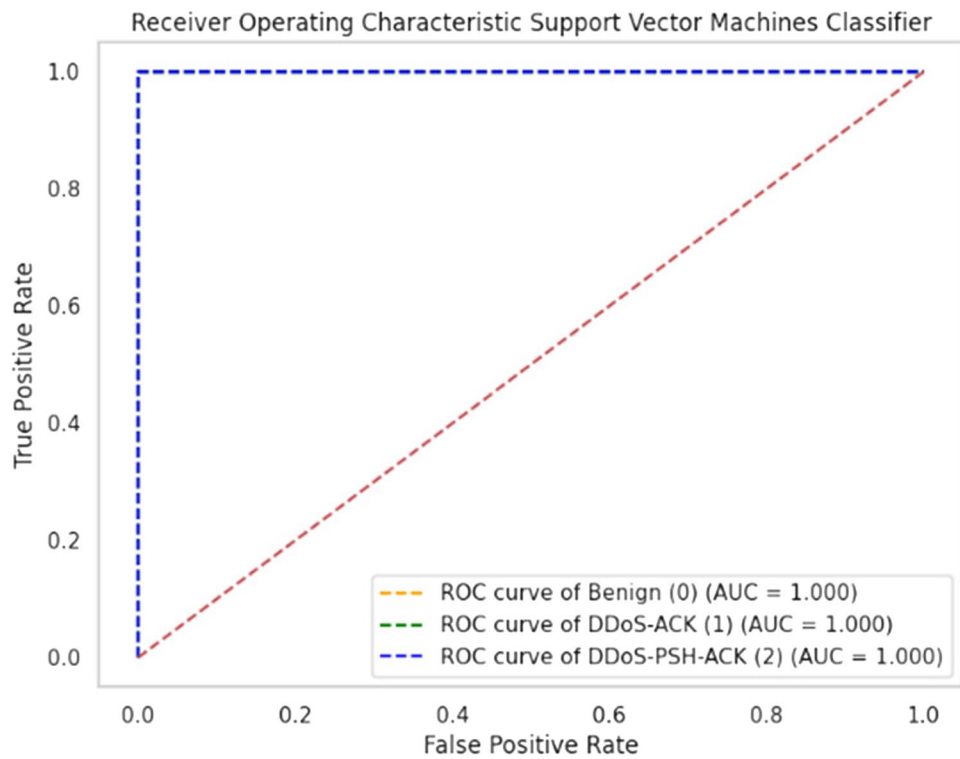


Fig. 9 Illustrates the Receiver Operating Characteristic Curve for Support Vector Machine

$$\begin{aligned}
 &\text{Regression of Nonlinear } (\text{Cost } (h\theta(x), y)) \\
 &= -\log(h\theta(x)), \text{ if } y = 1 \text{ and } (\text{Cost } (h\theta(x), y)) \quad (19) \\
 &= -\log(1 - h\theta(x)), \text{ if } y = 0
 \end{aligned}$$

$$\text{Regression of Nonlinear } Y = f(X, B) + \varepsilon \quad (20)$$

The following are the primary steps of MLP algorithms:

All algorithms can employ these scientific variables or techniques. This method is intended for overclassification and consists of Weight Determined, Stump Performance, Classifier's Significance, and Overall Error.

After the activation function at the hidden layer has yielded the necessary outcome, the derivatives that are partial with the required values should be extracted and delivered to a different layer within the Multi-Layer Perceptron.

The outcome is achieved up till steps two and three are repeated.

The estimates will be utilized in the last step to decide whether to adopt an option based on the outcomes for training data or a feed-forward method using MLP-specific activation approaches.

Finish. Through these steps, the MLP classifier determines the likelihoods of several class labels, enabling efficient classification according to the highest probability. To assign categories to previous data values and

make appropriate predictions, we employed the MLP approach. We used the SVM technique throughout the planning stage to make sure our forecasts fell into the right categories. The results of this method are shown in the resulting matrix, which is shown in Fig. 10.

ROC analysis, which gauges the classifier's effectiveness when the classification threshold is altered, can be used to assess the model's effectiveness. Because it takes into account both the costs and the benefits of decisions, this method is similar to cost-benefit analysis. The MLP classifier's receiver operating characteristic (ROC) curve is shown in Fig. 11. The curve provides details about the MLP's classification skills and shows how well it performs at different thresholds.

Adaboost (Adb) Method

An improvement method used as an ensemble method in machine learning-driven classification techniques is the AdaBoost process. To increase classifier accuracy, it incorporates several classifiers. The incremental ensemble method is called AdaBoost. To produce a potent classification system with exceptional performance, the AdaBoost classifier merges a large number of poorly performing classifiers. The fundamental idea behind AdaBoost is to ensure accurate predictions of rare events by building classifier weights and training data from a

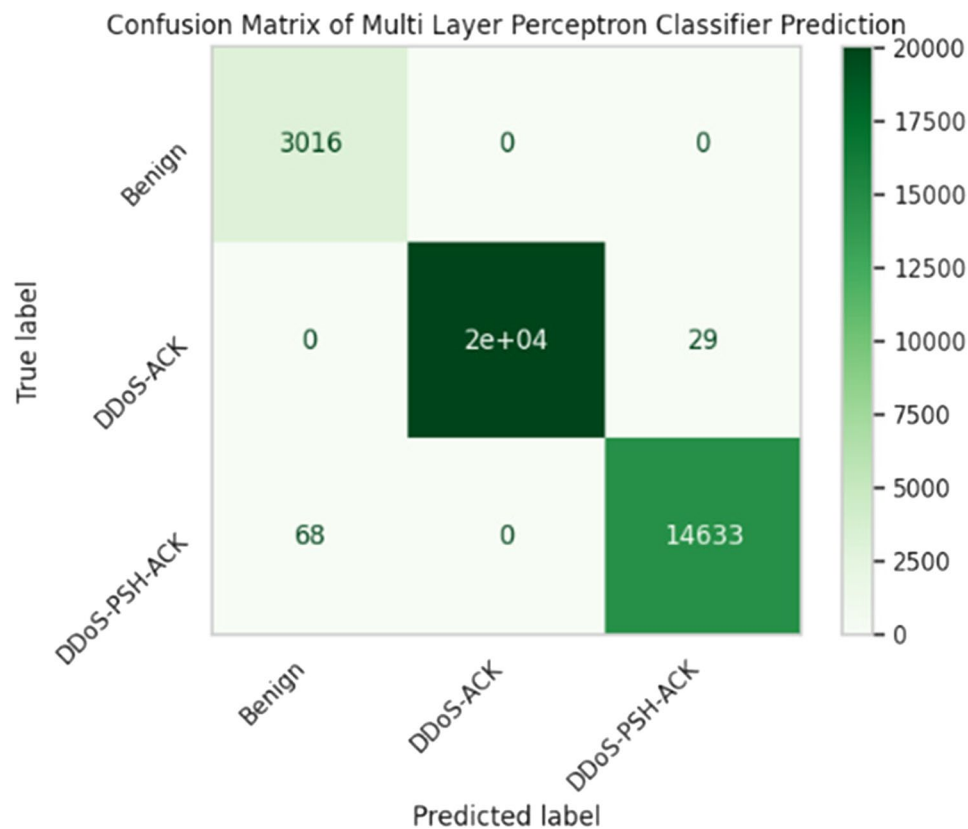


Fig. 10 Confusion Matrix of Multi-Layer Perceptron Algorithm

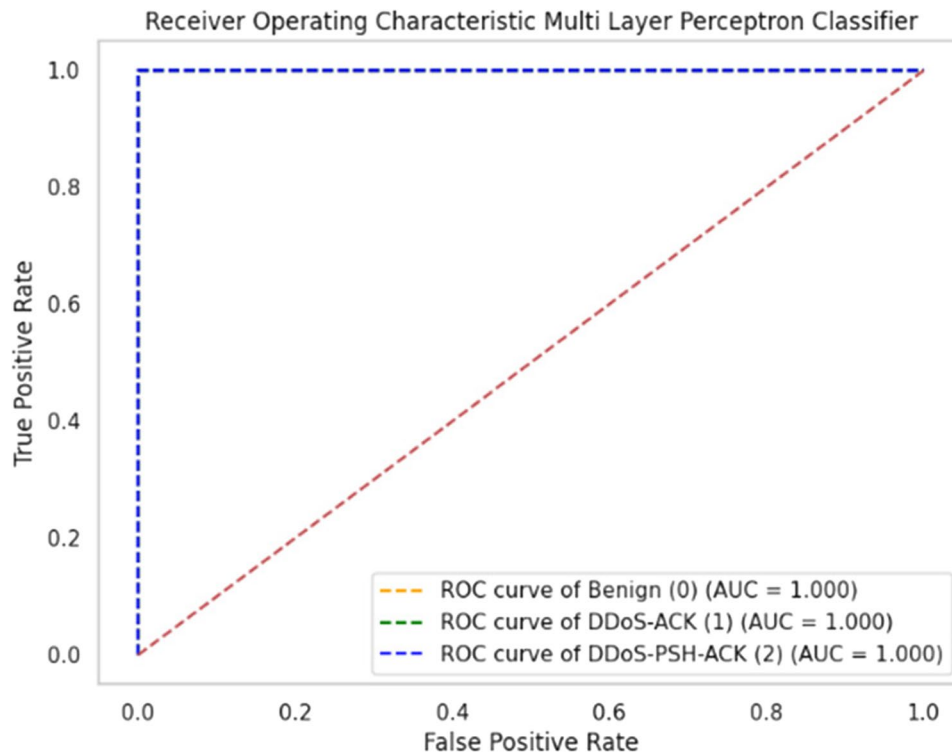


Fig. 11 Illustrates of the Receiver Operating Characteristic Curve for Matrix Multi-Layer Perceptron (MLP)

sample for every loop. The fundamental classifier can be any machine learning method that takes in a set of practice weights. Adaboost must meet two requirements: A range of weighted training samples should be used to actively train the machine learning model. These are some variables or scientific methods that can be applied to any algorithm. This method is based on classification and includes Stump Performance, Weight Determined, Classifier's Significance, and Total Error.

Weight Calculated

$$W(X_i, Y_i) = \frac{1}{N} \quad i = 1, 2, 3, \dots, n \quad (21)$$

$$\text{Performance of the} \quad \frac{1}{2} \log_e \frac{1 - \text{Total Error}}{\text{Total Error}} \text{Stump} \quad (22)$$

$$\text{Significance \& Total New sample weight} \\ = \text{old weight} * e^{\pm \text{Amount of say Error}} \quad (23)$$

The ADB algorithm adheres to a series of basic steps:

1. Adaboost chooses a training subset at random.
2. The training set is used to iteratively train the Adaboost learning model.
3. To increase their classification probability, incorrectly classified data are given more weight for the following iteration.

4. A higher weight will be assigned to the more accurate classifier. In each cycle, qualified classifiers are also given a weight based on their accuracy.
5. Until all training data is free of mistakes or achieves the maximum estimators, the process is repeated.
6. A "vote" including every learning algorithm is conducted for categorization.

The ADB method iteratively modifies the parameters to decrease the cost and optimize prediction. The old information has been labelled, and the ADB projection forecasts the value of the fresh data. Figure 12 displays the results of the ADB confusion matrix.

ROC evaluation is a useful tool for assessing the efficacy of a model for classifiers in altering the degree of discrimination threshold. This type of evaluation is very similar to the cost-benefit evaluation and aids in knowledgeable choices. Figure 13 shows an illustration of the receiver operating characteristic (ROC) curve for the ADB classifier, which illustrates the evaluation results while offering information about the classification's regular performance.

Experimental setup

Python is the primary programming language used in this study for data preprocessing, model implementation, and performance evaluation. The study utilizes Support Vector Machine (SVM), AdaBoost, Multi-Layer

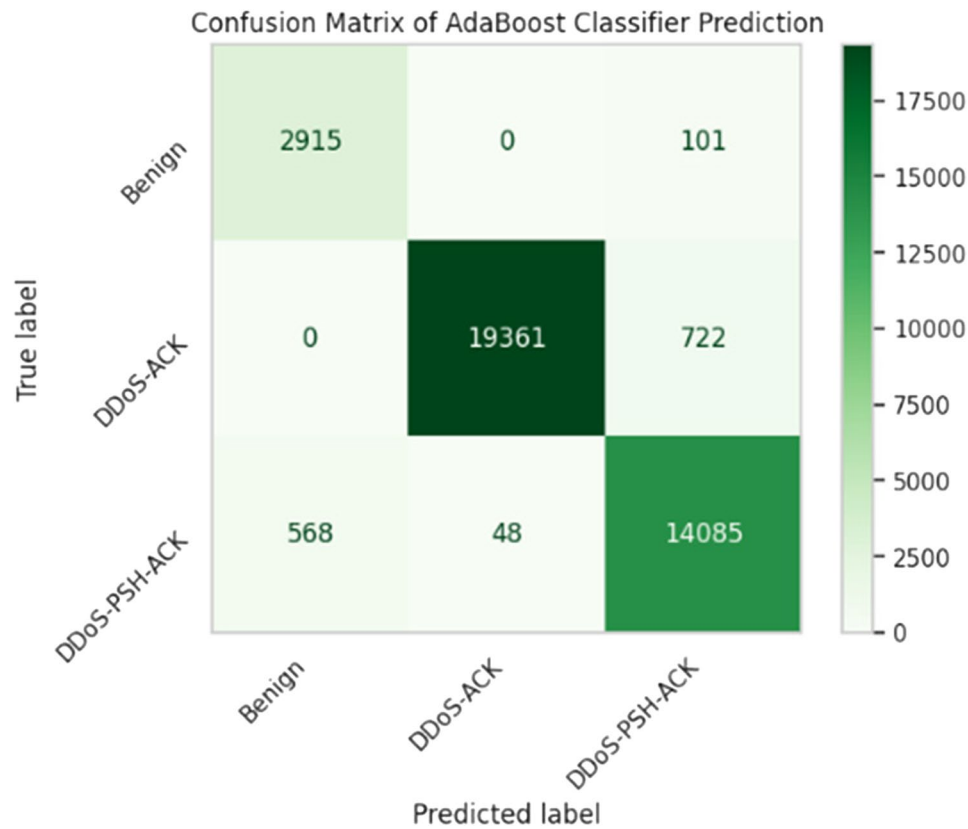


Fig. 12 Confusion Matrix AdaBoost (ADB) Algorithm

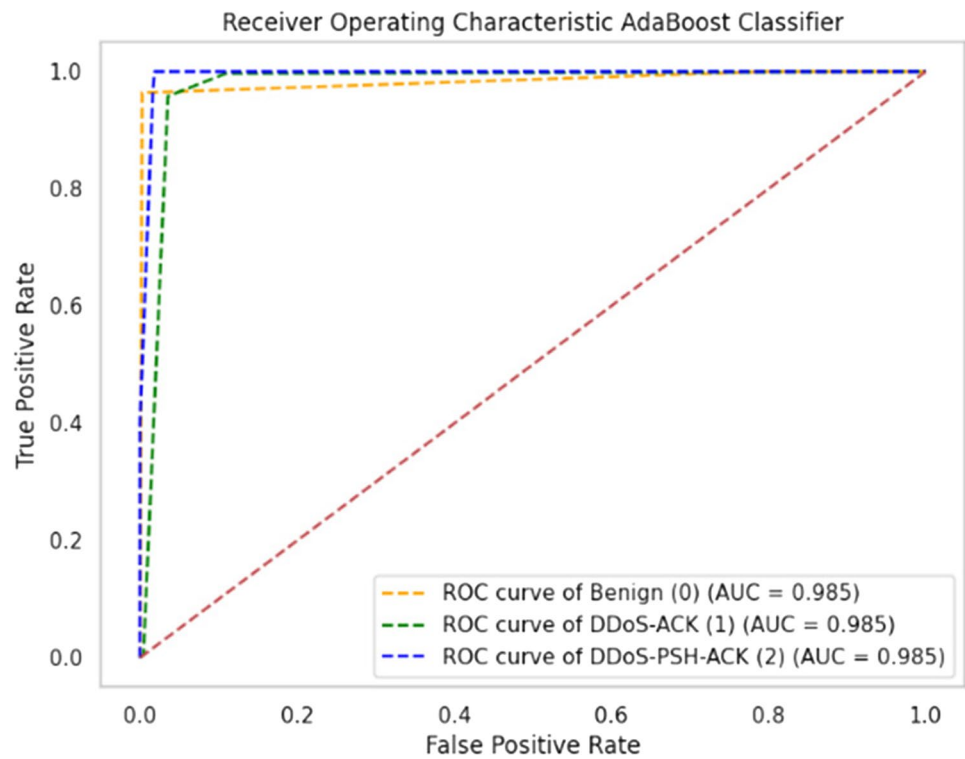


Fig. 13 Illustrates the Receiver Operating Characteristic (ROC) Curve for AdaBoost (ADB)

Table 6 Shows the accuracy of models that use a combination of algorithms for cyber-attack detection systems (CADS)

Hybrid Algorithm	Accuracy of Algorithms
Fuzzy C-Means, AdaBoost (ADB)	96.1931%
Proposed Method	
Fuzzy C-Means, Multi-Layer Perceptron (MLP) Proposed Method	99.7433%
Fuzzy C-Means, Support Vector Machines (SVM) Proposed Method	99.8597%

Perceptron (MLP), and Fuzzy C-Means clustering for cyber-attack detection. The development environment consists of Anaconda Navigator for managing dependencies and Jupyter Notebook as the coding interface.

The Cyber-Attacks Detection System (CADS) dataset, comprising 23 attributes and 151,200 records, was used for training and testing. The dataset underwent preprocessing to ensure data quality before being applied to three different model combinations: (1) SVM+ Fuzzy C-Means, (2) MLP+ Fuzzy C-Means, and (3) AdaBoost+ Fuzzy C-Means. Each combination was executed under the same experimental conditions to ensure a fair comparison.

All experiments were conducted on a personal computer with the following specifications: Intel Core i5-2520 M (2.5 GHz, 2nd generation), 4GB RAM, 500GB HDD, and Windows 10 (64-bit architecture). Software tools such as Scikit-learn, NumPy, and Jupyter Notebook were utilized for model implementation and evaluation.

This setup ensures a reproducible and structured approach to assessing machine learning techniques for cyber-attack detection.

Results and discussion

During the investigation, a variety of hybrid algorithms are thoroughly assessed to ascertain how well they forecast cyberattacks. The accuracy of hybrid algorithms is mentioned in the table below. Table 6 shows the accuracy of models and Table 7 illustrates the parameter scores for cyber-attack detection systems.

Results indicate that hybrid algorithms have demonstrated varying degrees of accuracy in predicting cyberattacks. The maximum accuracy of 99.8597% was

attained by combining fuzzy c means and support vector machines. According to these methods, combining these algorithms will produce a reliable model for predicting cyberattacks. Adaboost and Fuzzy C-C-Means received the 3rd-highest accuracy of 96.1931%, while Multi-Layer Perceptron, as well as Fuzzy C-Means combined, achieved the 2nd-greatest accuracy of 99.7433%.

The parameter values evaluating the combined algorithms' performance can be used to forecast cyberattacks. The precision value of 0.92641461, f1-value of 0.94298779, specificity value of 1.0, recall value of 0.96288640, and sensitivity value of 1.0 were attained by the combination of AdaBoost and fuzzy c means, whereas the recall value of 0.99797681, f1-value of 0.99494215, specificity value of 1.0, f1-value of 0.99494215, sensitivity value of 1.0, and precision value of 0.99199093 were obtained by the Multi-Layer Perceptron (MLP) and fuzzy C-Means. A recall value of 0.99858385, specificity value of 1.0, precision value of 0.99783636, sensitivity value of 1.0, and f1-value of 0.99820912 were attained by combining support vector machines and fuzzy c-means.

While this study demonstrates promising results across several machine learning models, it does not explicitly address the class imbalance present in the dataset, which may impact the sensitivity of minority class detection.

Model robustness

While the focus of this study is on classification performance under standard conditions, it is important to consider the resilience of the proposed model to adversarial threats. The hybrid approach, which combines Fuzzy C-Means clustering with SVM, naturally enhances robustness by introducing an unsupervised clustering step before supervised classification. This design helps dilute the effect of adversarial input perturbations near decision boundaries. Additionally, the SVM's margin-based classification is known to be relatively stable against small-scale adversarial manipulations. Future work will explore formal adversarial training strategies and ensemble defenses to further strengthen the model's security under adversarial conditions.

Table 7 The parameter scores for cyber-attack detection system (CADS) using different algorithm combinations

S/No.	Parameter Score	Fuzzy C-Means, AdaBoost (ADB) Algorithm	Fuzzy C-Means, Multi-Layer Perceptron (MLP) Algorithm	Fuzzy C-Means, Support Vector Machine (SVM) Algorithm
1	Precision	0.92641461	0.99199093	0.99783636
2	Recall	0.96288640	0.99797681	0.99858385
3	F1-Score	0.94298779	0.99494215	0.99820912
4	Sensitivity	1.0	1.0	1.0
5	Specificity	1.0	1.0	1.0

Model generalization and zero-day threats

The ability to generalize beyond known attack types is a critical aspect of cybersecurity systems. Our hybrid approach, which combines Fuzzy C-Means clustering (an unsupervised method) with SVM (a supervised classifier), enhances generalization by first grouping traffic patterns without relying on predefined labels. This allows the model to detect outlier or anomalous patterns, even if they do not match previously seen attack types. Although the current evaluation is based on labeled data, the clustering step gives the framework an inherent capability to identify suspicious, previously unseen traffic thus offering a level of resilience to zero-day attacks. In future work, this generalization can be further improved by integrating continual learning frameworks and synthetic data generation (e.g., via GANs) to simulate unknown threats and adapt the model dynamically to evolving attack vectors.

Discussion

The proposed hybrid machine learning model, which integrates Fuzzy C-Means clustering with Support Vector Machines (SVM), has shown significant improvements in the detection of DDoS attacks specifically DDOS-PSH-ACK and DDOS-ACK compared to conventional approaches. The inclusion of clustering in the preprocessing stage enabled more effective data segmentation, improving class separability and thereby enhancing SVM performance on imbalanced data. Experimental results demonstrate that the hybrid model consistently outperforms standalone classifiers such as AdaBoost, Multi-Layer Perceptron (MLP), and traditional SVM, particularly in terms of detection accuracy, precision, and the Matthews Correlation Coefficient (MCC). The high MCC values validate the robustness of the model under skewed class distributions—a common scenario in real-world attack detection datasets.

Compared to previous studies that often generalize attack detection across multiple types or rely on single classifiers, our method offers a targeted and refined

strategy for DDoS detection using a well-structured hybrid framework. Additionally, the use of multiple evaluation metrics enhances the comprehensiveness of our performance assessment. However, the study also has some limitations. First, although the APA-DDoS dataset provides rich labeled data, real-time adaptability and performance on unseen zero-day attacks remain challenges. Second, while clustering aids in feature space optimization, the computational complexity of hybrid models may limit scalability in large-scale or resource-constrained environments. Future work will focus on optimizing run-time performance and extending the model to incorporate adaptive learning for evolving threat landscapes.

Overall, the results confirm the potential of hybrid models in improving the accuracy and reliability of intrusion detection systems and encourage their broader adoption in IoT and cloud-based cybersecurity architectures.

Comparative analysis

Table 8 presents a comparative analysis of previous research results alongside the proposed methods for Cyber-Attack Detection Systems (CADS). The accuracy of different hybrid algorithms is evaluated to assess their effectiveness in detecting cyber threats. The study by Presekal et al. achieved an accuracy of 87.0%, while Alrumaih et al. reported a significantly higher accuracy of 98.62%. Qasim et al. further improved the detection accuracy to 99%.

In comparison, the proposed methods utilizing Fuzzy C-Means combined with different machine learning techniques demonstrated superior performance. The Fuzzy C-Means and AdaBoost (ADB) approach achieved an accuracy of 96.1931%. The combination of Fuzzy C-Means with Multi-Layer Perceptron (MLP) further improved accuracy to 99.7433%. The highest accuracy, 99.8597%, was achieved using Fuzzy C-Means with Support Vector Machines (SVM). These results indicate that the proposed hybrid models outperform previous research, demonstrating enhanced effectiveness in cyber-attack detection.

The increased accuracy of the proposed methods highlights their efficiency in identifying cyber threats more precisely than existing models. By integrating Fuzzy C-Means clustering with advanced machine learning algorithms, the proposed models improve detection rates while minimizing errors. The highest accuracy of 99.8597% using the Fuzzy C-Means and SVM method confirms the robustness and reliability of the approach. This improvement enhances the overall security framework, making the proposed methods more efficient for real-time cyber-attack detection.

Figures 14 and 15 show the accuracy of several algorithm pairings in the framework of mixed algorithms for

Table 8 Previous Research Results and Proposed Methods of the Cyber-attack Detection Systems (CADS)

Hybrid Algorithm	Accuracy of Algorithms
[44]	87.0%
[45]	98.62 %
[47]	99%
Fuzzy C-Means, AdaBoost (ADB)	96.1931
Proposed Method	%
Fuzzy C-Means, Multi-Layer Perceptron (MLP) Proposed Method	99.7433 %
Fuzzy C-Means, Support Vector Machines (SVM) Proposed Method	99.8597 %

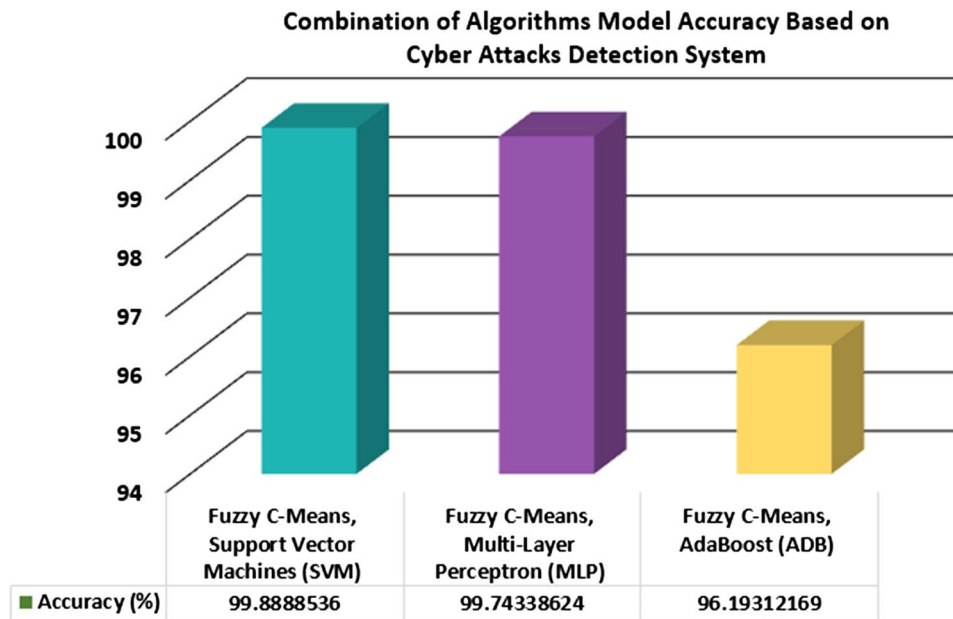


Fig. 14 Combination of Algorithms Model Accuracy Cyber-Attacks Detection System

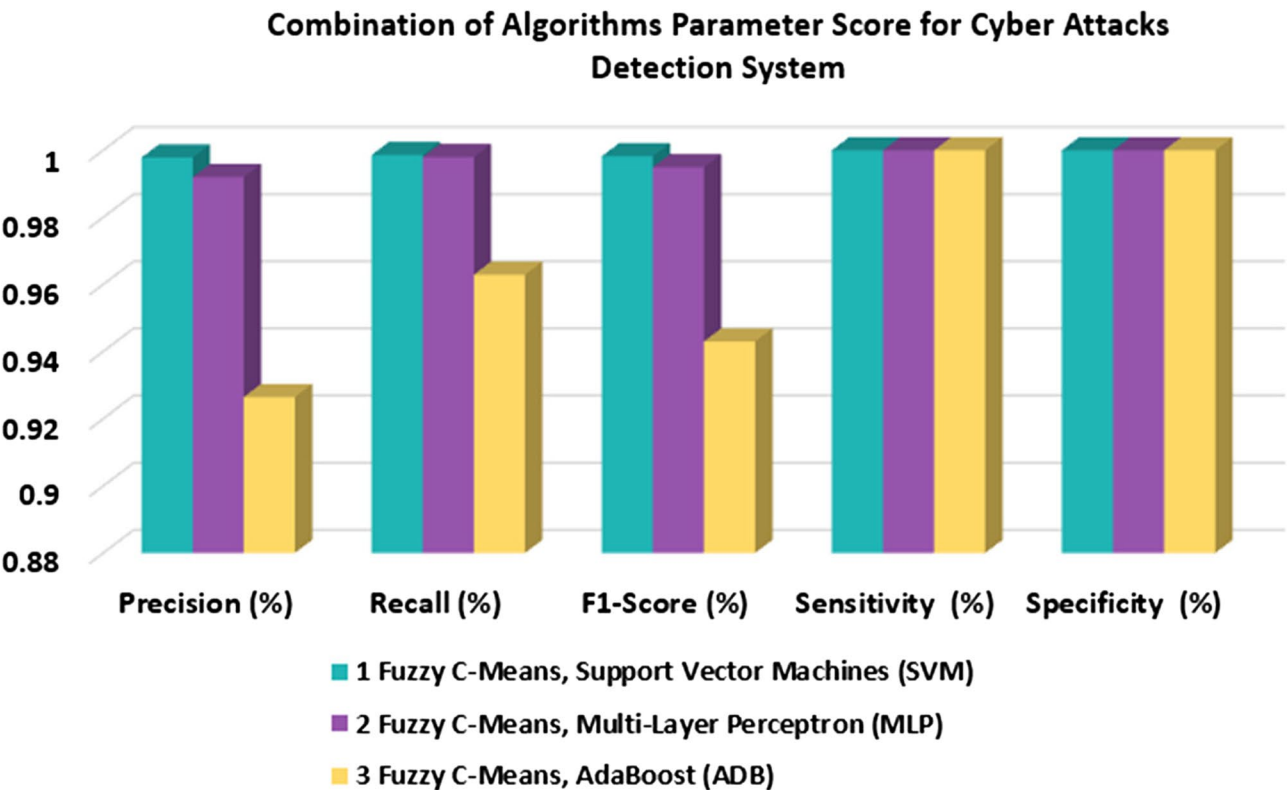


Fig. 15 Combination of methds Parameter Score Cyber-Attacks Detection System

cyberattack detection. The accuracy varies from 96.19 to 99.85%, represented by three bars. The maximum accuracy of 99.8597% was attained by combining fuzzy c means and support vector machines. While Adaboost and Fuzzy C-Means obtained an accuracy of 96.1933%,

Multi-Layer Perceptron and Fuzzy C-Means combined achieved a 99.7433% accuracy. The level of precision shown on the graph indicates that the combinations of methods have reached the highest level of prediction, based on the findings. For

example, characteristics such as recall, sensitivity, scoring precision, f1 score, and specificity have all reached optimal accuracy.

The proposed cyber-attack detection system (CADS) was validated using multiple techniques to ensure reliability and robustness. 10-fold cross-validation was applied to prevent overfitting and assess model stability. The dataset, obtained from Kaggle, consisted of 151,200 instances with 23 IoT-related attributes, including DDoS-PSH-ACK, DDoS-ACK, and Benign traffic. The dataset was split into 75% training and 25% testing to evaluate performance. Various hybrid models combining Fuzzy C-Means with AdaBoost (ADB), Multi-Layer Perceptron (MLP), and Support Vector Machines (SVM) were tested. The Fuzzy C-Means + SVM model achieved the highest accuracy of 99.8597%, outperforming previous studies, such as Presekal et al. (87.0%), Alrumaih et al. (98.62%), and Qasim et al. (99%). Performance metrics, including precision, recall, F1-score, sensitivity, and specificity, further validated the model's effectiveness. While the proposed model demonstrated superior accuracy, future work should explore additional real-world datasets and real-time deployment to enhance generalizability.

Conclusion

This study proposed a hybrid machine learning-based cybersecurity framework for detecting cyber threats, with a primary focus on Distributed Denial of Service (DDoS) attacks such as DDoS-ACK and DDoS-PSH-ACK, by integrating Fuzzy C-Means clustering with Support Vector Machines (SVM). The Fuzzy C-Means + SVM model achieved the highest detection accuracy of 99.8597%, surpassing other combinations such as Fuzzy C-Means + MLP (99.7433%) and Fuzzy C-Means + AdaBoost (96.1931%). Recursive Feature Elimination (RFE) was applied within cross-validation folds to prevent data leakage, ensuring optimal and reproducible feature selection. The framework demonstrated high precision, MCC robustness, and ROC/PR-AUC values, indicating strong performance in differentiating between malicious and benign traffic.

While promising, the study has limitations. Dataset diversity and class imbalance still affect generalizability, and no oversampling or class-weighting strategies were applied in this work. Computational complexity, including training and inference time, was not measured, which is critical for assessing real-world feasibility. Additionally, the absence of statistical significance testing (e.g., t-tests or ANOVA) means performance differences between models cannot yet be claimed as conclusive. The lack of explainability analysis (e.g., SHAP or LIME) and adversarial robustness validation (e.g., FGSM, PGD) are also important areas for future exploration.

Future directions

Future work will focus on expanding dataset diversity with real-world cyberattack traffic, applying class imbalance mitigation strategies such as SMOTE and class weighting, and evaluating statistical significance of model improvements. We plan to integrate explainability techniques to interpret model decisions, validate adversarial robustness, and design a system-level deployment architecture with a clear threat model. Additionally, exploring deep learning-based hybrid models and real-time intrusion detection mechanisms will be key for enhancing scalability and responsiveness. Robust evaluation will include MCC, ROC-AUC, and PR-AUC to ensure reliability in imbalanced scenarios, ultimately improving adaptability, precision, and resilience in next-generation Cyber-Attack Detection Systems (CADS).

Acknowledgements

Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R387), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Authors' contributions

Aun Abbas and Muqaddas Salahuddin perform the Original Writing Part, Software, and Methodology; Muhammad Zohaib Khan and Tehseen Mazhar perform the Rewriting, investigation, design Methodology, and Conceptualization; Abdullah Ayub Khan, Fahim Uz Zaman and Tehseen Mazhar perform related work part and manage results and discussions; Syed Azeem Inam, Muhammad Amir Khan and perform related work part and manage results and discussion; Aun Abbas Abdullah Ayub Khan, and Ghadah Aldehim perform Rewriting, design Methodology, and Visualization; Tehseen Mazhar and Ghadah Aldehim perform Rewriting, design Methodology, and Visualization.

Funding

Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R387), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Data availability

The data used to support the findings of this study are available from the corresponding authors upon request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interest

The authors declare no competing interests.

Received: 17 March 2025 / Accepted: 3 September 2025

Published online: 16 October 2025

References

1. Lambert II, GM. Security analytics: Using deep learning to detect cyber attacks. 2017.
2. Sarker IH. Machine learning: Algorithms, real-world applications and research directions. *SN computer science*. 2021;2(3):160.

3. Sarker IH. CyberLearning: Effectiveness analysis of machine learning security modeling to detect cyber-anomalies and multi-attacks. *Internet of Things*. 2021;14:100393.
4. Zeidanloo HR, ABA Manaf. Botnet detection by monitoring similar communication patterns. *arXiv preprint arXiv:1004.1232*. 2010.
5. Bhatia J, Sehgal R, Kumar S. Botnet command detection using virtual honeynet. *International Journal of Network Security & Its Applications*. 2011;3(5):177–189.
6. Stankovic S and Simic D. Defense strategies against modern botnets. *arXiv preprint arXiv:0906.3768*. 2009.
7. Rahim R, et al. Detecting the Phishing Attack Using Collaborative Approach and Secure Login through Dynamic Virtual Passwords. *Webology*. 2020;17(2).
8. Mishra N and Pandya S. Internet of things applications, security challenges, attacks, intrusion detection, and future visions: A systematic review. *IEEE Access*. 2021;9:59353–77.
9. Chen Z, et al. Machine learning-enabled iot security: Open issues and challenges under advanced persistent threats. *ACM Computing Surveys*. 2022;55(5):1–37.
10. Farraj A, Hammad E. A Physical-Layer Security Cooperative Framework for Mitigating Interference and Eavesdropping Attacks in Internet of Things Environments. *Sensors (Basel, Switzerland)*. 2024;24(16):5171.
11. Moubayed A. A Complete EDA and DL Pipeline for Software-defined 5G Network Intrusion Detection. *Future Internet*. 2024;16(9):331.
12. Kilichev D, Turimov D, and Kim W. Next-Generation Intrusion Detection for IoT EVCS: Integrating CNN, LSTM, and GRU Models. *Mathematics*. 2024;12(4):571.
13. Abdelhamid S, et al. Attention-Driven Transfer Learning Model for Improved IoT Intrusion Detection. *Big Data and Cognitive Computing*. 2024;8(9):116.
14. Chen J, Xiao J, and Xu J. VGGIncepNet: Enhancing Network Intrusion Detection and Network Security through Non-Image-to-Image Conversion and Deep Learning. *Electronics*. 2024;13(18):3639.
15. Aroon N, et al. An Architecture of Enhanced Profiling Assurance for IoT Networks. *Electronics*. 2024;13(14):2832.
16. Beshah YK, Abebe SL, and Melaku HM. Drift Adaptive Online DDoS Attack Detection Framework for IoT System. *Electronics*. 2024;13(6):1004.
17. Habibi O, Chemmakha M, and Lazaar M. Imbalanced tabular data modelization using CTGAN and machine learning to improve IoT Botnet attacks detection. *Engineering Applications of Artificial Intelligence*. 2023;118:105669.
18. Alabisi BA, Anbar M and Rihan SDA. Conditional tabular generative adversarial based intrusion detection system for detecting ddos and dos attacks on the internet of things networks. *Sensors*. 2023;23(12):5644.
19. Negera WG, et al. Lightweight Model for Botnet Attack Detection in Software Defined Network-Orchestrated IoT. *Applied Sciences*. 2023;13(8):4699.
20. Yang C, Guan W, and Fang Z. IoT botnet attack detection model based on DBO-catboost. *Applied Sciences*. 2023;13(12):7169.
21. de Caldas Filho FL, et al. Botnet detection and mitigation model for IoT networks using federated learning. *Sensors*. 2023;23(14):6305.
22. Maeng J, Heo Y, and Joe I. Hyperledger Fabric-Based Lightweight Group Management (H-LGM) for IoT Devices. *IEEE Access*. 2022;10:56401–9.
23. Fan S, Wang J. Multi-dimension-precision chaotic encryption mechanism for Internet of Things. *Internet of Things*. 2024;26:101202.
24. Bakhshi T, Ghita B and Kuzminykh I. A Review of IoT Firmware Vulnerabilities and Auditing Techniques. *Sensors*. 2024;24(2):708.
25. Al Hanif A, Ilyas M. Effective Feature Engineering Framework for Securing MQTT Protocol in IoT Environments. *Sensors*. 2024;24(6):1782.
26. SarhanM, et al. Feature extraction for machine learning-based intrusion detection in IoT networks. *Digital Communications and Networks*. 2024;10(1):205–16.
27. Sun H, et al. Power transmission lines foreign object intrusion detection method for drone aerial images based on improved yolov8 network. *Drones*. 2024;8(8):346.
28. Rajan DM, Aravindhar DJ. Detection and Mitigation of DDOS Attack in SDN Environment Using Hybrid CNN-LSTM. *Migration Letters*. 2023;20(S13):407–19.
29. Hu Z. Knowledge Graph Based Large Scale Network Security Threat Detection Techniques. *Applied Mathematics and Nonlinear Sciences*. 9(1).
30. Koroniotis N, et al. Towards developing network forensic mechanism for botnet activities in the IoT based on machine learning techniques. in *Mobile Networks and Management: 9th International Conference, MONAMI 2017, Melbourne, Australia, December 13–15, 2017, Proceedings 9*. 2018. Springer.
31. Mirsky Y, et al. Kitsune: an ensemble of autoencoders for online network intrusion detection. *arXiv preprint arXiv:1802.09089*. 2018.
32. Yuan X, Li C, Li X. DeepDefense: identifying DDoS attack via deep learning. in *2017 IEEE international conference on smart computing (SMARTCOMP)*. 2017. IEEE.
33. Pektaş A, Acarman T. Botnet detection based on network flow summary and deep learning. *International Journal of Network Management*. 2018;28(6):e2039.
34. Moon YH, et al. Detection of botnets before activation: an enhanced honeypot system for intentional infection and behavioral observation of malware. *Security and Communication Networks*. 2012;5(10):1094–1101.
35. Avci İ, Koca M. Cybersecurity Attack Detection Model, Using Machine Learning Techniques. *Acta Polytechnica Hungarica*. 2023;20(7):29–44.
36. Mahmood AM, Avci İ. Cybersecurity Defence Mechanism Against DDoS Attack with Explainability. *Mesopotamian Journal of CyberSecurity*. 2024;4(3):278–90.
37. Koca M, Avci I. A Novel Hybrid Model Detection of Security Vulnerabilities in Industrial Control Systems and IoT Using GCN+ LSTM. *IEEE Access*. 2024.
38. Vinayakumar R, et al. Deep learning approach for intelligent intrusion detection system. *Ieee Access*. 2019;7:41525–50.
39. Ramzan MS, et al. An Innovative Machine Learning based end-to-end Data Security Framework in Emerging Cloud Computing Databases and Integrated Paradigms: Analysis on Taxonomy, challenges, and Opportunities. *Spectrum of engineering sciences*. 2025;3(2):90–125.
40. Parveen F, et al. Real-Time Intrusion Detection with Deep Learning: Analyzing the UNR Intrusion Detection Dataset. *Journal of Computing & Biomedical Informatics*. 2024;7(02).
41. Jadhav AD, Pellakuri V. Highly accurate and efficient two phase-intrusion detection system (TP-IDS) using distributed processing of HADOOP and machine learning techniques. *Journal of Big Data*. 2021;8(1):131.
42. Maseno EM, Wang Z, Xing H. A systematic review on hybrid intrusion detection system. *Security and Communication Networks*. 2022;2022(1):9663052.
43. Lyu R, et al. Network intrusion detection based on an efficient neural architecture search. *Symmetry*. 2021;13(08):1453.
44. Okey OD, et al. BoostedEnML: Efficient technique for detecting cyberattacks in IoT systems using boosted ensemble machine learning. *Sensors*. 2022;22(19):7409.
45. Presekal A, et al. Attack graph model for cyber-physical power systems using hybrid deep learning. *IEEE Transactions on Smart Grid*. 2023;14(5):4007–20.
46. Alrumaih TN, Alenazi MJ. CGAAD: Centrality-and Graph-Aware Deep Learning Model for Detecting Cyberattacks Targeting Industrial Control Systems in Critical Infrastructure. *IEEE Internet of Things Journal*. 2024.
47. Dini P, Saponara S. Analysis, design, and comparison of machine-learning techniques for networking intrusion detection. *Designs*. 2021;5(1):9.
48. Quigley K, Burns C, Stallard K. 'Cyber Gurus': A rhetorical analysis of the language of cybersecurity specialists and the implications for security policy and critical infrastructure protection. *Government Information Quarterly*. 2015;32(2):108–17.
49. salman Qasim S, NSAIF SM. Advancements in time series-based detection systems for distributed denial-of-service (ddos) attacks: A comprehensive review. *Babylonian Journal of Networking*. 2024;2024:9–17.
50. Mazhar T, et al. Analysis of IoT security challenges and its solutions using artificial intelligence. *Brain Sciences*. 2023;13(4):683.
51. Ghadi YY, et al. Machine learning solution for the security of wireless sensor network. *IEEE Access*. 2024.
52. Ghadi YY, et al. Security risk models against attacks in smart grid using big data and artificial intelligence. *PeerJ Computer Science*. 2024;10:e1840.
53. Verma R, Chandra S. RepuTE: A soft voting ensemble learning framework for reputation-based attack detection in fog-IoT milieu. *Engineering Applications of Artificial Intelligence*. 2023;118:105670.
54. Kaur J, Agrawal A, Khan RA. P2ADF: a privacy-preserving attack detection framework in fog-IoT environment. *International Journal of Information Security*. 2023;22(4):749–62.
55. Waheed N, et al. Fedblockhealth: A synergistic approach to privacy and security in iot-enabled healthcare through federated learning and blockchain. in *GLOBECOM 2023-2023 IEEE Global Communications Conference*. 2023. IEEE.
56. Rodríguez E, et al. Transfer-learning-based intrusion detection framework in IoT networks. *Sensors*. 2022;22(15):5621.
57. Khan NW, et al. A hybrid deep learning-based intrusion detection system for IoT networks. *Math. Biosci. Eng.* 2023;20(8):13491–520.

58. Zaman S, et al. Thinking out of the blocks: Holochain for distributed security in IoT healthcare. *IEEE Access*. 2022;10:37064–81.
59. Nandanwar H, Katarya R. Deep learning enabled intrusion detection system for Industrial IOT environment. *Expert Systems with Applications*. 2024;249:123808.
60. Nandanwar H, Katarya R. TL-BILSTM IoT: transfer learning model for prediction of intrusion detection system in IoT environment. *International Journal of Information Security*. 2024;23(2):1251–77.
61. Nandanwar H, Katarya R. Securing Industry 5.0: An explainable deep learning model for intrusion detection in cyber-physical systems. *Computers and Electrical Engineering*. 2025;123:110161.
62. Kauhsik B, Nandanwar H, Katarya R. Iot security: a deep learning-based approach for intrusion detection and prevention. in 2023 International Conference on Evolutionary Algorithms and Soft Computing Techniques (EASCT). 2023. IEEE.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.